UNDEF

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Carrera Lic. en Ciberdefensa

Álgebra I

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MÁS SOBRE SISTEMAS LINEALES:

EJ.:

a) $\begin{cases} x_1 + x_2 - x_3 = 2 \\ 2x_1 - x_2 + 2x_3 = 7 \\ 4x_1 + 2x_2 + x_3 = 3 \end{cases} \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -1 & 2 & 7 \\ 4 & 2 & 1 & 3 \end{array}\right] \xrightarrow[F_3 - 4F_1]{F_2 - 2F_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -3 & 4 & 3 \\ 0 & -2 & 5 & -5 \end{array}\right] \longrightarrow$

$\xrightarrow{-\frac{1}{3}F_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -\frac{4}{3} & -1 \\ 0 & -2 & 5 & -5 \end{array}\right] \xrightarrow{F_3 + 2F_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -\frac{4}{3} & -1 \\ 0 & 0 & \frac{7}{3} & -7 \end{array}\right]$ $\quad rg(A) = rg(A|b) = 3$

$\begin{cases} x_1 + x_2 - x_3 = 2 \\ x_2 - \frac{4}{3}x_3 = -1 \\ \frac{7}{3}x_3 = -7 \end{cases}$

$\frac{7}{3}x_3 = -7 \longrightarrow \boxed{x_3 = -3}$

$x_2 - \frac{4}{3}(-3) = -1$

$x_2 + 4 = -1$

$\boxed{x_2 = -5}$

$x_1 - 5 - (-3) = 2$

$\boxed{x_1 = 4}$

$\therefore \boxed{S: \vec{X} = (4, -5, -3)}$

(S.C.D.)

b) $\begin{cases} x_1 + 2x_2 - 4x_3 = 1 \\ 2x_1 + 5x_2 - 6x_3 = 0 \\ -x_1 - 3x_2 + 2x_3 = 3 \end{cases}$

c) $\begin{cases} x_1 + 2x_2 - 4x_3 = 1 \\ 2x_1 + 5x_2 - 6x_3 = 0 \\ -x_1 - 3x_2 + 2x_3 = 1 \end{cases}$

d) $\begin{cases} x_1 + 2x_2 - 4x_3 = 0 \\ 2x_1 + 5x_2 - 6x_3 = 0 \\ -x_1 - 3x_2 + 2x_3 = 0 \end{cases}$

$\left[\begin{array}{ccc|c|c} 1 & 2 & -4 & 1 & 1 \\ 2 & 5 & -6 & 0 & 0 \\ -1 & -3 & 2 & 3 & 1 \end{array}\right] \xrightarrow[F_3 + F_1]{F_2 - 2F_1} \left[\begin{array}{ccc|c|c} 1 & 2 & -4 & 1 & 1 \\ 0 & 1 & 2 & -2 & -2 \\ 0 & -1 & -2 & 4 & 2 \end{array}\right] \xrightarrow{F_3 + F_2} \underbrace{\left[\begin{array}{ccc|c|c} 1 & -2 & -4 & 1 & 1 \\ 0 & 1 & 2 & -2 & -2 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right]}_{b)} \quad c)$

b) $\left[\begin{array}{ccc|c} 1 & -2 & -4 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 2 \end{array}\right]$ $\quad rg(A) = 2 < rg(A|b) = 3$

$\hookrightarrow \boxed{S.I.} \Rightarrow \boxed{S = \emptyset}$

$0 = 2$ FALSO

c)

$$\left[\begin{array}{ccc|c} 1 & -2 & -4 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right] \qquad rg(A) = rg(A|b) = 2 < \#(\text{INCÓGNITAS}) = 3$$

→ SISTEMA COMPATIBLE INDETERMINADO

$$\begin{cases} x_1 - 2x_2 - 4x_3 = 1 \\ x_2 + 2x_3 = -2 \longrightarrow x_2 = -2x_3 - 2 \end{cases}$$

$$\rightarrow x_1 = -2x_2 + 4x_3 + 1$$

$$x_1 = -2(-2x_3 - 2) + 4x_3 + 1$$

$$x_1 = 8x_3 + 5$$

$$\therefore S: X = (8x_3 + 5, -2x_3 - 2, x_3)$$

$$\boxed{S: X = \alpha(8, -2, 1) + (5, -2, 0)} \qquad (S.C.I.)$$

d)

$$\left[\begin{array}{ccc|c} 1 & -2 & -4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \qquad rg(A) = rg(A|b) = 2 < 3 = \#(\text{INCÓGNITAS})$$

$$\Longrightarrow \boxed{S.C.I.}$$

$$\begin{cases} x_1 - 2x_2 - 4x_3 = 0 \\ x_2 + 2x_3 = 0 \longrightarrow x_2 = -2x_3 \end{cases}$$

$$\rightarrow x_1 = -2x_2 + 4x_3$$

$$x_1 = -2(-2x_3) + 4x_3$$

$$x_1 = 8x_3$$

$$\therefore S: X = (8x_3, -2x_3, x_3)$$

$$\boxed{S: X = \alpha(8, -2, 1)} \qquad (S.C.I.)$$